



IILM UNIVERSITY, GREATER NOIDA

## Internship Presentation

# Prompt Engineering Research and Integration

### STUDENT NAME

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### PROGRAM & SEMESTER

B.Tech CSE | 3rd Semester | Section 2CSE9

### ENROLLMENT NUMBER

25SCS1003004568

### ORGANIZATION

Excelerate (RIT Tiger STRIPES Program)

## INTERNSHIP OVERVIEW

# A Remote Research Internship in Applied Prompt Engineering

Over four structured weekly phases, this internship applied prompt engineering principles to AI-assisted digital learning platforms — researching core techniques, designing and testing structured prompt templates across five real-world use cases, and evaluating them against a multi-criteria performance rubric.



*Powered by Excelerate, supported by Rochester Institute of Technology's Tiger STRIPES Program.*

**Jun 08 – Jul 08,  
2026**

Internship Duration

**5**

AI Use Cases Designed

**18**

Documented Test Runs

**1.5 → 4.7 / 5**

Overall Score Improvement

# What This Internship Set Out to Achieve

01

## Research

Study established prompting techniques and their applicability to educational AI systems.

02

## Design

Design structured prompt templates for five representative AI learning-platform use cases.

03

## Iterate

Evolve every prompt from a vague draft (V-A) to a schema-compliant final version (V-C).

04

## Evaluate

Score each version against a 5-criteria rubric using a standardised evaluation matrix.

05

## Consolidate

Package findings into a production-ready prompt suite with phased recommendations.

# Four Core Problems in AI-Assisted Learning Platforms

01

## Inconsistent Content Quality

Unstructured prompts cause factual errors and reading-level mismatches — risking learner trust (Bender et al., 2021).

02

## Limited Personalisation

Generic, one-size-fits-all content undermines autonomy and motivation, per Self-Determination Theory (Deci & Ryan, 1985).

03

## Poorly Structured Pathways

Missing prerequisite logic causes cognitive overload, violating Ausubel's Assimilation Theory (1968).

04

## Inconsistent Opportunity Descriptions

Ad hoc drafting produces tonal inconsistency and heavy editorial overhead (Cornelissen, 2020).

# The Six-Technique Prompt Engineering Toolkit

1

## Zero-Shot

Direct instruction, no examples — fast but sensitive to ambiguity.

2

## Few-Shot

2–5 example input-output pairs anchor the model's expected pattern.

3

## Chain-of-Thought

Step-by-step reasoning before the final answer — improves multi-step accuracy.

4

## Role / Persona

An assigned expert identity shapes tone, depth and register.

5

## Constraint-Based

Explicit rules for format, length and reading level remove ambiguity.

6

## Iterative Refinement

Draft → test → analyse → revise, treated as an empirical loop.

# Five AI Use Cases Designed and Tested

01

**Adaptive Quiz Generator**

*Bloom's-aligned MCQs with instant feedback*

02

**Concept Explanation Generator**

*6-part scaffolded explanations*

03

**Structured Learning Pathway Generator**

*Prerequisite-aware, week-by-week pathways*

04

**Programme Description Generator**

*Brand-consistent scholarship/opportunity copy*

05

**Badge & Visual Identity Generator**

*Micro-credential briefs for design teams*

# Adaptive Quiz Generator

*Bloom's-aligned MCQs with structured JSON output and instant feedback*

## FINAL PROMPT STRUCTURE (V-C)

**ROLE:** Expert educational assessor with mastery of Bloom's Taxonomy

**CONSTRAINTS:** Bloom's level, MCQ format, target reading grade

**OUTPUT:** JSON: question, options, answer, explanation, bloom\_level, difficulty

**TECHNIQUE:** Role + Few-Shot + Structured Output

## EXPECTED BENEFIT

Personalised, calibrated questions with immediate, targeted feedback — sustaining challenge without frustration.

## SUCCESS KPI

Knowledge retention (pre/post) and engagement time-on-task.

# Concept Explanation Generator

*Six-part scaffolded explanations tuned to a learner's age and prior knowledge*

## FINAL PROMPT STRUCTURE (V-C)

**ROLE:** Patient, encouraging tutor for a specified age group

**STRUCTURE:** Definition → Analogy → Steps → Example → Misconception → Check

**CONSTRAINTS:** Reading grade, 250–350 word length, encouraging tone

**TECHNIQUE:** Role + Chain-of-Thought + Constraint

## EXPECTED BENEFIT

Scaffolded explanations reduce cognitive load; the misconception flag pre-empts common errors.

## SUCCESS KPI

Comprehension quiz scores and learner satisfaction rating.



# Structured Learning Pathway Generator

*Prerequisite-aware, week-by-week pathways aligned to Bloom's Taxonomy*

## FINAL PROMPT STRUCTURE (V-C)

**ROLE:** Senior curriculum designer in competency-based education

**REASONING:** 4-step CoT: sub-competencies → order → Bloom's map → weekly plan

**OUTPUT:** JSON array: week, topic, bloom\_level, prerequisites, milestone

**TECHNIQUE:** Role + Chain-of-Thought + Structured JSON

## EXPECTED BENEFIT

Pedagogically sound, scaffolded pathways replace ad hoc topic lists  
— cutting design time ~60%.

## SUCCESS KPI

Pathway completion rate; prerequisite alignment audit score.

# Programme / Opportunity Description Generator

*Brand-consistent scholarship and opportunity copy in a fixed 5-section structure*

## FINAL PROMPT STRUCTURE (V-C)

**ROLE:** Professional communications writer for the institution

**STRUCTURE:** Summary → Who Should Apply → What You Gain → How to Apply → Key Dates

**CONSTRAINTS:** 150–220 words; self-validation checklist before output

**TECHNIQUE:** Role + Format Constraint + Self-Validation

## EXPECTED BENEFIT

Zero missing fields and consistent brand voice — editorial review time cut from ~45 to ~5 minutes.

## SUCCESS KPI

Editorial revision rate; time-to-publish.

# Badge & Visual Identity Content Generator

*Structured, design-ready briefs for micro-credential badges*

## FINAL PROMPT STRUCTURE (V-C)

**ROLE:** UX designer and brand strategist for educational credentials

**STRUCTURE:** Skill Statement → Evidence → Visual Metaphor → Colour → Accessibility

**CONSTRAINTS:** Anti-cliché instruction; verifiable skill statement

**TECHNIQUE:** Role + Multi-Section Structured Output

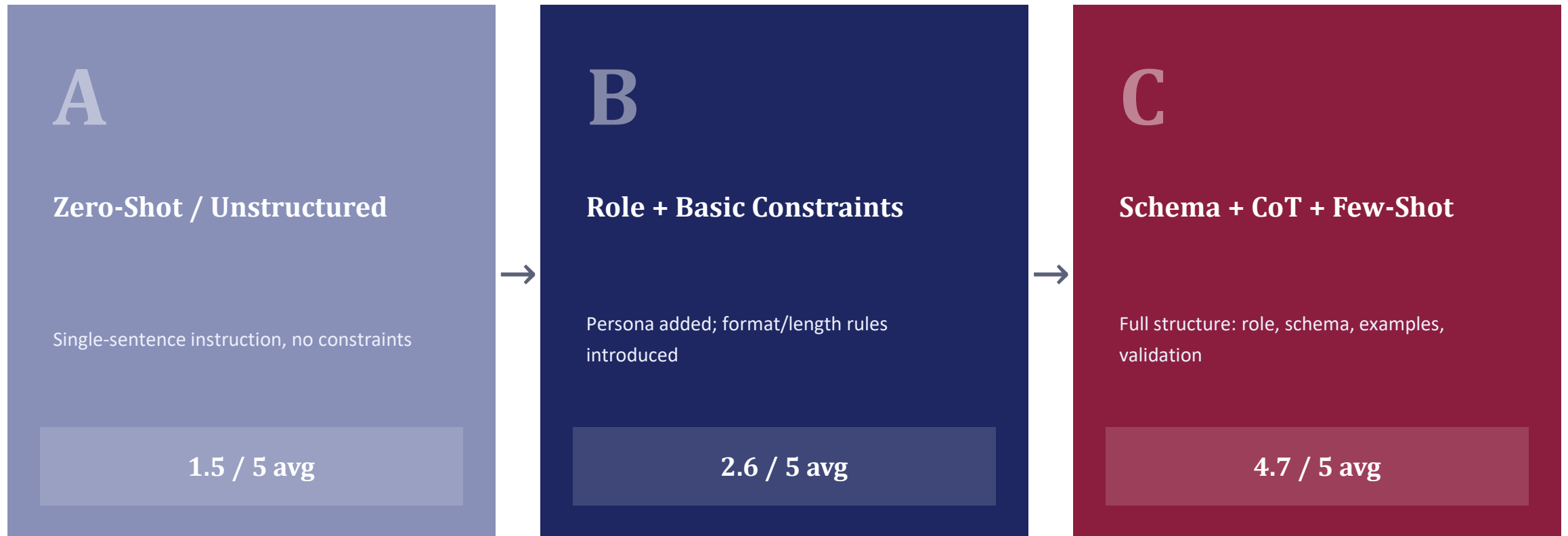
## EXPECTED BENEFIT

Accelerates the badge design pipeline with clear, verifiable skill descriptors that support learner motivation.

## SUCCESS KPI

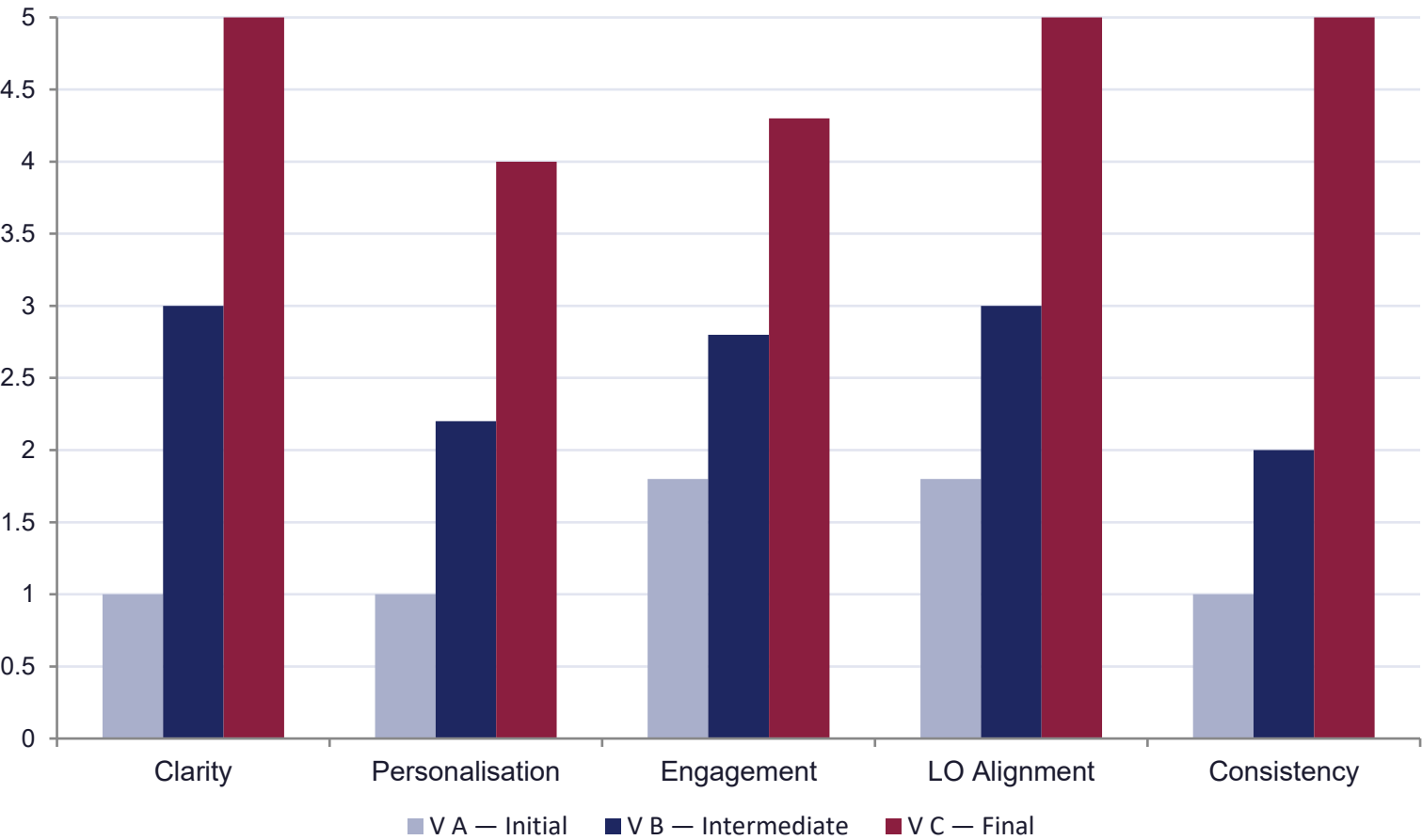
Badge claim rate; design brief-to-final iteration count.

# An Empirical, Iterative Prompt Evolution



*Applied consistently across all 5 use cases — 18 documented test runs, scored on Clarity, Personalisation, Engagement, LO Alignment & Consistency.*

# Score Improvement Across Prompt Versions



OVERALL AVERAGE

1.5 → 4.7

+3.2 points on a 5-point scale, V-A to V-C

KEY READ

Consistency & LO Alignment gained the most (schema-driven); Personalisation lagged — it needs runtime learner data, not just a better prompt.

# Prompting Strategies, Ranked by Measured Impact

| # | Strategy                       | Impact      | Why It Works                                                         |
|---|--------------------------------|-------------|----------------------------------------------------------------------|
| 1 | Output Schema / Format Spec    | Highest     | Eliminates format variance; enables programmatic parsing             |
| 2 | Role / Persona Prompting       | High        | Highest effort-to-impact ratio; one sentence changes tone & register |
| 3 | Chain-of-Thought Reasoning     | High        | Externalises planning; critical for multi-step tasks like pathways   |
| 4 | Bloom's Taxonomy Tagging       | High        | Ensures pedagogical alignment; prevents difficulty mismatches        |
| 5 | Few-Shot Examples              | Medium–High | Reduces ambiguity for complex, nested output formats                 |
| 6 | Self-Validation Instruction    | Medium      | Lightweight quality gate; catches ~20% of omission errors            |
| 7 | Word Count / Length Constraint | Medium      | Prevents verbose output; aids engagement and readability             |

# Expected Outcomes and Deployment Recommendations

## KEY OUTCOMES

- Validated, reusable prompt suite — all 5 use cases scoring 4.4+ / 5
- Schema specification + role prompting confirmed as highest-ROI techniques
- Personalisation identified as the ceiling for prompt-only solutions
- A ranked, documented framework to guide future prompt design work

## PHASED ROADMAP FOR EXCELERATE

### IMMEDIATE

Deploy Programme Description & Quiz Generator prompts — highest ROI, ~85% editorial time saved.

### SHORT-TERM

Integrate few-shot libraries and self-validation checks across remaining use cases.

### STRATEGIC

Build runtime learner-profile variables to close the personalisation gap; add governance review.



# Thank You

Abhishek Raj Bharti | Prompt Engineering Research Intern | Excelerate

*Questions & Discussion*